



Loops That Make Patterns

Ontario Math C3.1 — repeating events

Name: _____

Date: _____

★ I can use a repeat loop to make a pattern that repeats.

Patterns repeat

Look at a striped shirt or a bead bracelet — the same part shows up again and again. That is a pattern.

A repeat loop runs ALL the blocks inside it each time around. So if you put two stamps inside, the loop makes a repeating pattern for you. Watch Bit make a bead pattern.

Read this loop

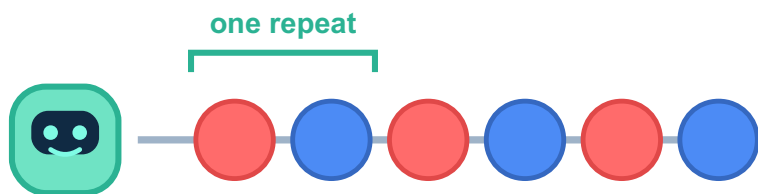
when green flag clicked

repeat 3

stamp red bead

stamp blue bead

Both stamps run each time. 3 repeats of (red, blue) makes 6 beads: red, blue, red, blue, red, blue.



repeat 3 times → 6 beads

Bit stamps red, blue — and the loop repeats it 3 times.

1 Count it

The loop repeats 3 times and stamps 2 beads each time. How many beads each repeat? _____ How many beads in TOTAL? _____

2 Predict, then check

Change repeat 3 to repeat 4. PREDICT how many beads Bit makes now, then CHECK by counting 2 for each repeat. How many beads?

3 Write your own

Make Bit stamp a heart-then-star pattern 5 times. Write the repeat number: repeat _____ → (stamp heart, stamp star)

4 Challenge

Now there are THREE stamps inside: repeat 3 → (●, ▲, ■). How many stamps in total? Draw what the pattern looks like.

Big idea

A loop repeats the WHOLE inside each time. The blocks inside are one piece of the pattern — repeat them to make the pattern grow.



Loops That Make Patterns

Quick facilitation notes & answer key

What this worksheet teaches

A repeat loop runs EVERY block inside it each time, so a loop with two or more actions builds a repeating pattern. Students count, change, and write pattern loops. No coding experience needed.

Before you start

No computer needed — the worksheet shows the blocks and what Bit does, so you can teach from the printout. Optional: a Scratch-style app to drag the blocks and watch Bit run — you only press the green flag.

How to lead it (about 30 minutes)

1. Show it (you do). The loop stamps 2 beads each time; 3 repeats make 6 beads.
2. Do one together. Exercise 1 — 2 beads each repeat, 6 in total.
3. Let them work. Students predict the change (Ex 2), write a repeat (Ex 3), and count a three-stamp loop (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — 2 beads each repeat; 6 beads total (2×3). Exercise 2 — 8 beads (2×4); predict 8.

Exercise 3 — repeat 5 (makes 10 stamps). Exercise 4 (challenge) — 9 stamps (3×3); the pattern is heart-star-square, three times.

If students get stuck — and ways to adjust

Watch for: counting only one block per repeat. Every block inside runs each time.

Make it easier: circle the blocks inside the loop, then multiply by the repeat number.

Make it harder: ask the total for a four-stamp loop repeated 3 times (12).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The child multiplies the blocks inside by the repeat number to find the total.